

Flow Matching

Discussion #10: Solutions

Written by:

Justin, Konpat

Conventions used throughout (“flow” formulation)

- $x_0 \sim \mathcal{N}(0, I)$ is noise, $x_1 \sim p_{\text{data}}$ is clean data, and $t \in [0, 1]$ goes from noise ($t = 0$) to clean ($t = 1$).
- Noisy point: $x_t = t x_1 + (1 - t) x_0$. Conditional velocity: $\frac{dx_t}{dt} = x_1 - x_0$ (constant along the straight path).
- The network is trained with $\mathcal{L} = \mathbb{E}_{t, x_0, x_1} \|(x_1 - x_0) - u_\theta(x_t, t)\|^2$. The minimizer is the *marginal flow*

$$u(x, t) = \mathbb{E}[x_1 - x_0 \mid x_t = x] = \frac{\mathbb{E}[x_1 \mid x_t = x] - x}{1 - t}.$$
- For a dataset of points $\{d_i\}$ with frequencies π_i , the posterior weights are $w_i(x, t) \propto \pi_i \exp\left(-\frac{\|x - t d_i\|^2}{2(1-t)^2}\right)$ and $\mathbb{E}[x_1 \mid x_t = x] = \sum_i w_i d_i$.

1 Warmup: Noise Shapes

The noise x_0 always has *exactly the shape of one data sample*, because x_t is an element-wise interpolation between the two.

0) Generating a grayscale 1920×1080 image.

$x_0 \in \mathbb{R}^{1 \times 1080 \times 1920}$, i.e. shape $(1, 1080, 1920)$ in (C, H, W) order (PyTorch convention). $(1080, 1920)$ is also correct if the channel dimension is dropped. With a batch of B images: $(B, 1, 1080, 1920)$. Note that height comes before width.

1) Generating a random point (x, y) within a square.

$x_0 \in \mathbb{R}^2$, shape $(2,)$. (Batched: $(B, 2)$.)

2 Visualizing Flow in 2D

Data: $A = (-1, 0)$ and $B = (1, 0)$, each with probability $\frac{1}{2}$. Figure 1 collects the answers to 1.0–1.5.

1.0 Sketch the contour of $\mathcal{N}(0, I)$ at 1 standard deviation.

A circle of radius 1 centered at the origin. It passes exactly through A and B .

Side note: in 2D only $1 - e^{-1/2} \approx 39\%$ of the Gaussian mass lies inside this circle ($\approx 86\%$ inside radius 2), so plenty of noise samples land outside it.

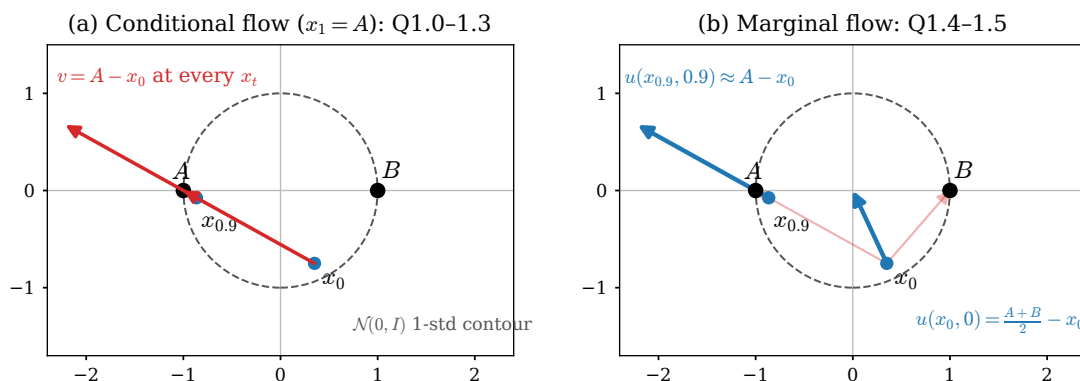


Figure 1: (a) Conditional flow for one training pair $(x_0, x_1 = A)$. (b) Marginal flow at x_0 and $x_{0.9}$. Faint red arrows in (b) are the two conditional velocities $A - x_0$ and $B - x_0$ whose average is the blue arrow at x_0 .

1.1 Pick one “random” x_0 and plot it.

Any point is acceptable, preferably near the high-probability region. We use $x_0 = (0.35, -0.75)$.

1.2 Say $x_1 = A$; plot $x_{0.9}$.

$x_{0.9} = 0.9A + 0.1x_0 = (-0.9, 0) + (0.035, -0.075) = (-0.865, -0.075)$.
It lies on the straight segment from x_0 to A , 90% of the way to A .

1.3 For all points x_t , draw the velocity vector $x_{\text{clean}} - x_{\text{noise}}$.

$v = A - x_0 = (-1.35, 0.75)$, and it is the **same vector at every point** on the path (at x_0 , at $x_{0.9}$, and everywhere in between), because x_t is linear in t . Drawn to scale (one unit of time), the arrow from x_0 ends exactly at A , while the identical arrow drawn from $x_{0.9}$ extends well past A .

1.4 For the point x_0 , plot the marginal flow (roughly).

At $t = 0$, $x_t = x_0$ is pure noise and carries no information about which data point it will be paired with, so $w_A = w_B = \frac{1}{2}$:

$$u(x_0, 0) = \frac{1}{2}(A - x_0) + \frac{1}{2}(B - x_0) = \frac{A+B}{2} - x_0 = -x_0 = (-0.35, 0.75).$$

The arrow points from x_0 to the **origin**, the mean of the dataset. Note that the origin is *not* a data point.

1.5 For the point $x_{0.9}$, plot the marginal flow (roughly).

At $t = 0.9$ the noise has standard deviation $1 - t = 0.1$, so $x_{0.9} = (-0.865, -0.075)$ is only consistent with A :

$$\frac{w_B}{w_A} = \exp\left(-\frac{\|x_{0.9} - 0.9B\|^2 - \|x_{0.9} - 0.9A\|^2}{2(0.1)^2}\right) = \exp\left(-\frac{3.121 - 0.007}{0.02}\right) \approx e^{-156} \approx 10^{-68}.$$

So $u(x_{0.9}, 0.9) \approx A - x_0 = (-1.35, 0.75)$: the marginal flow **equals the conditional flow** toward A . It is a long vector; the *remaining* displacement $(1 - t)u = 0.1u$ is what takes $x_{0.9}$ exactly onto A .

1.6 In English, explain the direction of the marginal flow at $t = 0$ and $t = 0.9$.

The marginal flow always points from x_t toward the network's best guess of the clean sample, $\mathbb{E}[x_1 | x_t]$.

- $t = 0$: the noise says nothing about the data, so the best guess is the average of the whole dataset. Every arrow points toward the dataset mean (here the origin). The model “hedges” between A and B .
- $t = 0.9$: x_t is mostly signal, so only one data point is plausible and the flow points straight at it (here A). That point is the one whose scaled position td is closest to x_t (weighted by how frequent it is in the data), which is usually, but not always, the nearest data point.

As t goes from 0 to 1, the field changes from “everything flows to the mean” to “everything flows to its own data point” (Figure 2).

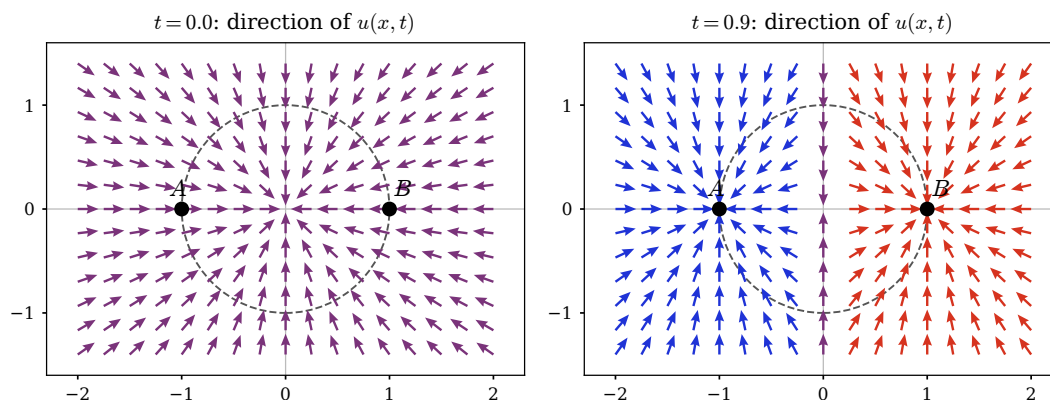


Figure 2: Direction of the marginal flow everywhere in the plane. Blue arrows are heading to A , red to B , purple are undecided. At $t = 0$ all arrows point to the mean; at $t = 0.9$ the plane has split into two basins, one per data point.

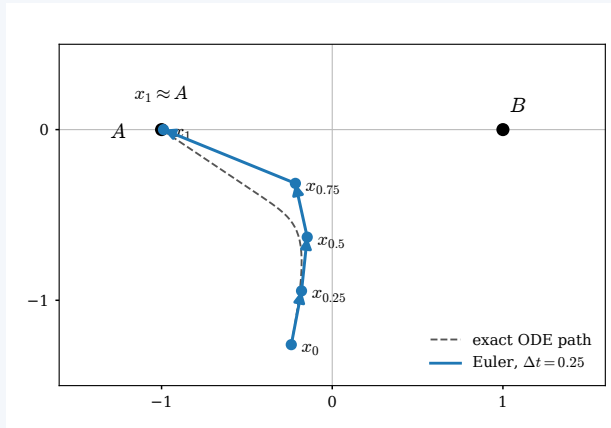
1.7 Start from x_0 , draw the flow sampling procedure with 4 sampling steps.

Use Euler steps of size $\Delta t = 0.25$, following the marginal flow at each step:

$$x_{t+\Delta t} = x_t + \Delta t u(x_t, t).$$

Starting from $x_0 \approx (-0.24, -1.26)$ (read off the worksheet figure) and using the exact marginal flow:

t	x_t	$w_A(x_t, t)$	where u points
0.00	$(-0.24, -1.26)$	0.50	dataset mean (origin)
0.25	$(-0.18, -0.95)$	0.54	almost the mean, slight lean to A
0.50	$(-0.15, -0.63)$	0.64	between the mean and A
0.75	$(-0.22, -0.32)$	0.99	A
1.00	$(-0.99, 0.00)$		(sample $\approx A$)



What to notice:

- The first step heads toward the **mean**, not toward A . The sample only “commits” to A late, once x_t is far enough from the y -axis relative to the remaining noise.
- The y -coordinate shrinks linearly, $y_t = (1 - t)y_0$, because both data points have $y = 0$.
- The **last step lands exactly on the denoised guess**: with $\Delta t = 1 - t$, $x_1 = x_t + (1 - t)\frac{\mathbb{E}[x_1|x_t] - x_t}{1 - t} = \mathbb{E}[x_1 | x_{0.75}] \approx (-0.99, 0)$.
- The exact ODE path (dashed) is curved, so 4 straight Euler steps only approximate it. More steps follow it more closely.

1.8 What if I wanted to generate A twice as often as B ? How could I modify the standard training pipeline?

Change the **data distribution seen during training**: when sampling the clean point d for each training example, pick A with probability $\frac{2}{3}$ and B with probability $\frac{1}{3}$. Equivalent options: duplicate A in the dataset (e.g. $\{A, A, B\}$), or keep uniform sampling but give A -examples twice the weight in the loss. The noise, the interpolation, and the sampler stay unchanged.

Why it works: the posterior weights now include the prior, $w_d \propto \pi_d \mathcal{N}(x_t; t d, (1 - t)^2 I)$, so

- at $t = 0$ the flow points to the new mean $\frac{2}{3}A + \frac{1}{3}B = (-\frac{1}{3}, 0)$ instead of the origin;
- the boundary in noise space between “ends at A ” and “ends at B ” moves from $x = 0$ to $x \approx 0.43 = \Phi^{-1}(\frac{2}{3})$, so exactly $\frac{2}{3}$ of the Gaussian noise flows to A (Figure 3).

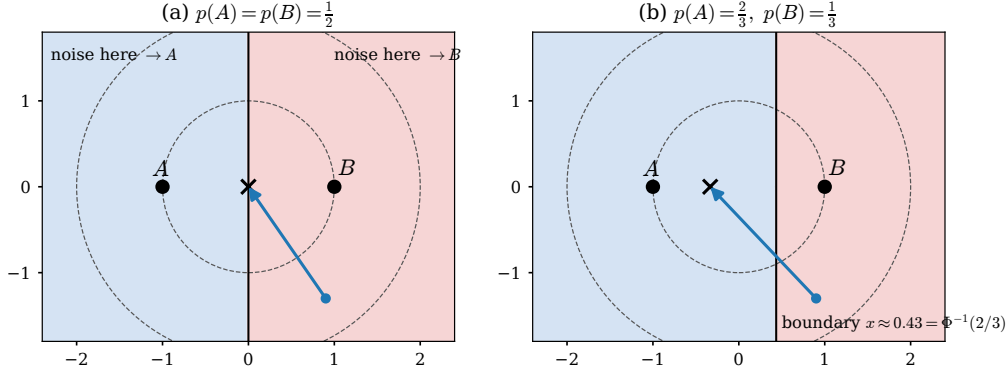


Figure 3: Where each starting noise x_0 ends up after following the marginal flow to $t = 1$. Blue: ends at A; red: ends at B. The $\times$ marks the dataset mean, which the $t = 0$ flow (blue arrow) points to.

3 Conditioning and Guidance

Data are spread uniformly along the three sides of an equilateral triangle centered at the origin: **red** = left side, **green** = right side, **blue** = bottom side. We use $x_0 \approx (-0.45, -1.35)$, as on the worksheet. At $t = 0$ the posterior equals the prior, so each flow points from x_0 to the *mean* of the data it was trained on:

$$m_{\text{red}} = (-0.45, 0.26), \quad m_{\text{green}} = (0.45, 0.26), \quad m_{\text{blue}} = (0, -0.52), \quad m_{\varnothing} = (0, 0) \text{ (centroid)}.$$

3.1 Explain how one could sample from a specific color without additional model input by changing training.

Train **three separate models**, each only on the data of one side: u_{red} on red points, u_{green} on green points, u_{blue} on blue points. To get a red sample, run the usual sampler with u_{red} ; its flow can only lead to the red side.

Drawbacks, and why conditioning is preferred in practice: you need one model per condition, each sees only a fraction of the data, and this cannot scale to open-ended conditions like text prompts. A single network $u_{\theta}(x, t, c)$ that takes c as input does the same job with shared weights.

3.2 Prompted flow: draw where $u_{\theta}(x, t, \text{red})$ and $u_{\theta}(x, t, \text{green})$ point.

At x_0 : $u(x_0, 0, \text{red}) = m_{\text{red}} - x_0 = (0, 1.61)$, pointing to the **midpoint of the red side**; $u(x_0, 0, \text{green}) = m_{\text{green}} - x_0 = (0.9, 1.61)$, pointing to the **midpoint of the green side**. (Figure 4a.)

3.3 Unprompted flow: draw where $u_{\theta}(x_0, t, \varnothing)$ points.

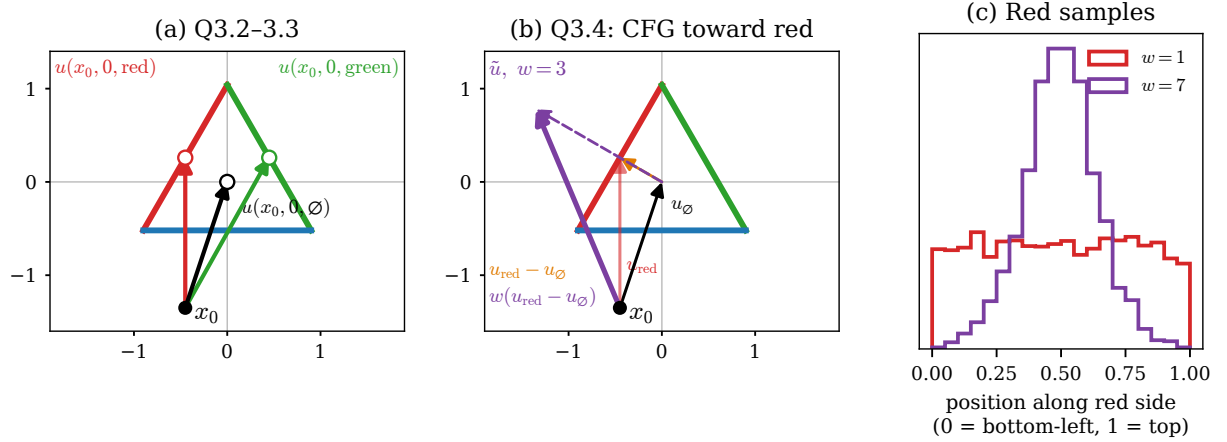


Figure 4: (a) Prompted flows toward red and green, and the unprompted flow toward the centroid (open circles mark the means). (b) Building the CFG vector for red with $w = 3$. (c) Where 4000 red samples land along the red side, without guidance ($w = 1$) and with strong guidance ($w = 7$).

$u(x_0, 0, \emptyset) = m_\emptyset - x_0 = (0.45, 1.35)$, pointing to the **center of the triangle**, the mean of all the data. (In practice the $\emptyset$ model is the same network, trained with the condition randomly dropped, e.g. 10% of the time.)

3.4 Guidance: draw the CFG vector $\tilde{u}_\theta(x, t, c) = u_\theta(x, t, \emptyset) + w(u_\theta(x, t, c) - u_\theta(x, t, \emptyset))$ for a large w . How does it differ from the standard prompted vector?

For $c = \text{red}$ at x_0 :

$$u_{\text{red}} - u_\emptyset = m_{\text{red}} - m_\emptyset = (-0.45, 0.26),$$

the direction that separates “red” from “average data”. With $w = 3$:

$$\tilde{u} = (0.45, 1.35) + 3(-0.45, 0.26) = (-0.90, 2.13).$$

How it differs from the prompted vector $u_{\text{red}} = (0, 1.61)$ (Figure 4b):

- $w = 0$ gives the unconditional flow and $w = 1$ gives exactly the prompted flow. For $w > 1$ the vector **extrapolates past** u_{red} along $u_{\text{red}} - u_\emptyset$: it is longer and rotated further away from the other classes, so at $t = 0$ it overshoots the red side.
- Effect on samples: they become “more red than red”. Guided samples avoid the parts of the red side that look like other classes (the corners shared with green and blue) and pile up in the middle of the red side (Figure 4c).
- Trade-off: better prompt adherence and more “typical” samples, but **less diversity**. For very large w , samples can be pushed off the data manifold (in images: oversaturated, unnatural results).